Supplementary Methods and Analyses

The supplement documents production refinements, revision diagnostics, the retrospective temporal assessment and subsequent biological interpretation. Frozen temporal outcomes are distinguished from the later human review (S6.6) and contemporary GO classification (S8).

S1. Optional post-hoc shortlist refinement

The production scoring and tiering procedure starts from 20,116 Ensembl 113 protein-coding IDs, yields 20,081 analysis labels after 35 duplicate-symbol consolidations (including 573 unresolved ENSG fallback labels), and ends at the 62-gene HIGH tier. Remaining names derive from the Ensembl GTF gene_name field or an exact-ID legacy-cache fallback; these sources do not validate current canonical HGNC symbols. Table S1 contains the 35 merged-symbol mappings (34 MANE Select mappings and one fallback); Table S2 contains 1,694 excluded labels (1,666 below the LOW threshold and 28 with NULL composite scores). The analyses below are optional, post-hoc annotations evaluated within HIGH and do not alter composite scores, priority tiers, or the six production evidence layers.

S1.1 Tau tissue-specificity thresholds

Tau specificity was propagated from tissue_expression into the scored-gene output and evaluated only as an exploratory annotation after production tier assignment. The final HIGH tier contains 62 genes. Threshold-specific Tau counts are reported in the generated expression-shortlist artifacts rather than treated as a production result. Duplicate tissue-expression records are aggregated before thresholding, and missing Tau is retained as missing.

S1.2 GSE135913 fetal cochlear data

Processed cluster-level matrices from GEO series GSE135913 [41] were used in local-cache-only mode for human cochlea samples GSM4037819 (15 weeks), GSM4037820 (17 weeks), and GSM4037821 (23 weeks). The local input relative paths and SHA-256 checksums are recorded in data/report/exploration/gse135913_hair_cell_provenance.json; this manifest contains no source URLs and should not be read as complete source-level provenance or proof of immutable inputs for the broader production run. The aggregate output is restricted to 15,608 of the 20,081 current analysis labels.

The matrices provide standardized CellFindR mean expression by cluster but do not include cell-type labels. Putative hair-cell clusters were therefore identified with a prespecified panel that excludes all Usher and SYSCILIA positive-control genes: *ATOH1*, *OTOF*, *POU4F3*, *GFI1*, *RBM24*, *LMO7*, *EPS8L2*, and *SLC17A8*. A sample was accepted only when at least four markers were represented, the highest aggregate marker mean was positive, and its margin over the next-ranked cluster was at least 0.5. The 15-week sample selected cluster 11 (8/8 markers; margin 4.735) and the 17-week sample selected cluster 10 (8/8; margin 5.221). The 23-week sample was excluded: five markers were represented, but the margin between the first and second clusters was only 0.097.

To test marker dependence, cluster selection was repeated after removing each marker individually. Every leave-one-marker-out run selected cluster 11 at 15 weeks and cluster 10 at 17 weeks. Because the primary marker panel contains neither *MYO7A* nor *PCDH15*, retention of these or other Usher controls is not used to define the clusters. In the final HIGH tier, 61 of 62 genes had a non-NULL GSE135913 aggregate; this exploratory resource is not part of the production expression layer.

Per-gene standardized means from accepted clusters were averaged across samples with NULL preservation. The output also records contributing sample count and a developmental-consistency measure. In the final HIGH tier, the fetal hair-cell Q75 strategy retained 16 genes and retained two of the two established Usher controls present in HIGH, but only two of the three total positive controls present in HIGH. Hair-cell Q75

denotes the 75th percentile calculated among genes already assigned to HIGH; it is a relative experimental-priority annotation, not a claim of absent expression below the threshold.

S1.3 Interpretation

Tau, hair-cell Q75, photoreceptor Q75, HPA-retina concordance, and the hair-cell/photoreceptor intersection represent alternative biological assumptions. In the final rerun, photoreceptor Q75 retained 16 genes, the hair-cell/photoreceptor intersection retained five, and HPA-retina concordance was NA/not evaluable because the local HPA-retina threshold was unavailable. Direct protein evidence retained 23 genes, the expression-OR-protein strategy retained 34, and the expression-AND-protein strategy retained five. They are not a sequential filtering cascade. Missing or below-threshold expression does not change the production tier and is not treated as evidence against disease relevance.

S2. Data-driven weight fitting

As post-hoc internal robustness analyses, we evaluated cross-validated grid search over the weight simplex and L2-penalized logistic regression using the selected disease controls. The grid search evaluated 8,856 candidate weight vectors. Logistic regression produced clipped-and-normalized weights [0, 0, 0, 0.032, 0.903, 0.065] for gnomAD, expression, annotation, localization, animal model, and literature, respectively. It changed the pooled known-control median from 92.8% to 98.7% and the housekeeping median from 94.8% to 75.1%. The default raw composite ≥0.70 housekeeping count was 2, whereas the logistic-regression count was 0; the default gated HIGH count was already 0. The default-versus-logistic-regression full-ranking Spearman correlation was 0.646. In a five-fold positive-label-held-out internal cross-validation summary, the held-out positive median was 98.2%; the matched negatives were constructed globally rather than held out within each fold, so leakage is possible and this is not independent held-out validation. These analyses concentrated weight on the animal-model layer and substantially altered the genome-wide ranking. Because the same limited control set would influence model selection and evaluation, they were treated as tests of possible over-specialization, not as independent validation or a basis for replacing the biologically specified production weights. The production sensitivity analysis used absolute ±0.05 and ±0.10 weight-point perturbations: each raw delta was applied to one weight and then all six weights were renormalized to sum to one, so the final change is not literally the raw delta and the other weights also change. Sixteen of 24 met $\rho \geq 0.85$, with mean $\rho = 0.8548$ and range 0.6051–0.9770. Implementations and detailed outputs are provided by scripts/weight_tuning.py and scripts/weight_logreg.py.

S3. Implemented evidence semantics and source coverage

The raw composite is the sum of observed weighted layer values divided by the sum of observed weights. Weights are constraint/expression 0.20 each and annotation/localization/animal/literature 0.15 each. These are heuristic priorities, not fitted probabilities. An observed zero participates in the denominator; NULL does not. The legacy output column confidence_tier is retained for schema compatibility but denotes a priority tier. HIGH requires score ≥0.70, at least three observed layers and the source-level gate described in the main Methods. MEDIUM requires score ≥0.40 and at least two layers; LOW requires score ≥0.20. The gate is not applied to raw ranks.

S3.1 Expression

To include neural expression context in the cilia-focused panel, we used cerebellum, which had measurements in both HPA and GTEx. This was a pragmatic proxy choice; it does not provide cochlear evidence.

Within each quantitative source, target means (retina/cerebellum) are divided by background means (testis/fallopian tube) only when both exist and the background is positive. HPA quantitative measurements take precedence over the ordinal HPA fallback when target and background columns are available in the source table. For ordinal HPA fallback, categories Not detected/Low/Medium/High are coded 0/1/2/3, then

normalized within tissue using average ranks among positive observations; zero stays zero and NULL stays NULL. Target/background contrast is computed on these tissue percentiles, never on the raw ordinal codes. Available HPA and GTEx contrasts are averaged. Their resulting positive contrast values are percentile-normalized before combination.

Quantitative restricted-panel Tau is $\Sigma(1-x_i/\max(x))/(n-1)$, requiring at least two observed tissues and a positive maximum, calculated separately for HPA nTPM and GTEx TPM. Available source Taus are averaged. This is specificity within the four-tissue panel, not whole-body specificity. Census photoreceptor expression is an independently ranked target feature. Quantitative/ordinal target values are normalized within their respective source columns, with positive average ranks divided by the positive-observation count and observed zeros retained as zero. The maximum available target percentile supplies the third component. Expression combines contrast percentile (0.4), source Tau (0.3) and maximum target percentile (0.3), renormalizing over non-NULL components. A gene may therefore receive partial expression evidence from background-only quantitative Tau even with no measured retinal target. No hair-cell component enters production scoring.

The stored 20,116-ID table has 0 HPA quantitative-retina observations, 0 GTEx-retina observations, 112 retinal HPA protein observations and 20,078 non-NULL Census photoreceptor values including zeros. The production proxy-removal sensitivity removes cerebellum columns and recomputes source transforms and genome-wide ranks rather than simply subtracting a component. It retains the original remaining measurements and weights. Per-control source values, expression contribution and resulting rank/tier changes are in the expression diagnostic tables in Additional file 11.

S3.2 Annotation, localization, animal and literature

Annotation uses $0.5 \times \log_2(\text{GO count}+1)/\text{maximum } \log_2(\text{GO count}+1)$, 0.3 × UniProt score/5 and 0.2 × binary pathway membership. Missing subcomponents contribute zero; all three missing yields NULL. This differs from the between-layer available-weight denominator.

Localization proximity checks direct compartment flags first (1.0), recognized adjacent compartments second (0.5), embedded compendium fallback third (0.3), then observed non-ciliary location (0). Adjacent strings include cytoskeleton, microtubule, cell junction and focal adhesion. HPA reliability factors are Enhanced/Supported=1.0 and Approved/Uncertain=0.6; all are antibody-staining modalities. If HPA reliability is unavailable, curated compendium evidence has factor 0.5. Mixed evidence uses the HPA factor rather than adding two scores. No usable modality yields NULL. Embedded lists have no validated historical snapshot date and are not an acquired complete experimental/proteomics atlas. The HIGH gate ORs the five direct compartment flags and two compendium flags, or animal score ≥0.12765713253595054. A positive adjacent localization score alone is insufficient. has_cilia_signal instead records positive aggregate localization or animal evidence and is not equivalent to gate passage.

Animal evidence unions distinct MGI/IMPC mouse term names and adds the zebrafish count to obtain a combined sensory count. The base score is 0.4 × mouse phenotype-presence flag × mouse HCOP confidence + 0.3 × zebrafish phenotype-presence flag × zebrafish confidence (HIGH=1.0, MEDIUM=0.7, LOW=0.4). It is multiplied by $\log_2(\text{combined sensory count}+1)/\log_2(\text{global maximum combined count}+1)$, then clipped to [0,1]; counts are not separately log-normalized per species. The absence of an ortholog is NULL; an observed mapping without recorded sensory phenotype can be zero. Human curation and animal experiments may share published knowledge.

Literature context counts have weights cilia=2, sensory=2, cytoskeleton=1 and polarity=1; missing contextual subcounts contribute zero. Implemented classification checks direct experimental first, then high-throughput, functional, incidental and none, with factors 1.0/0.3/0.6/0.1/0.0. Direct classification requires one shared PMID in the gene, experimental and cilia/sensory context sets. The quality-weighted count is divided by $\log_2(\text{total gene-linked publications}+1)$. Positive raw values receive their average rank among all non-NULL raw values

divided by the non-NULL count; nonpositive values are zero and missing raw scores remain NULL. This does not fully correct ascertainment, nor does “direct” establish clinical causality.

S4. Control traces, completeness, correlations and ranking sensitivities

The control and robustness diagnostics were specified before their outcomes. Additional file 11 contains the protocol, definitions and complete tables. The nine established Usher and 28 selected SYSCILIA controls are distinct evaluation groups, not a random validation cohort. Percentiles use (minimum tied rank−1)/(n−1) among non-NULL scores; top-K positions break ties by retained Ensembl ID.

All 62 HIGH labels have five or six layers (33 and 29), with zero supported by only three or four. Raising the minimum to four/five changes none; six retains 29. Eight raw top-100 labels have only one layer, all classified LOW. Layer count correlates with composite score at ρ=0.2401835 (n=20,053). Pairwise layer correlations use pair-specific observed genes and retain zeros. Annotation–literature is the largest (ρ=0.370354; n=19,114); correlations are descriptive and do not demonstrate independent information sources.

Each of 24 absolute ±0.05/±0.10 perturbations changes one raw weight, then renormalizes all weights to one. Together with equal weights these form a finite 25-configuration family. Membership is reported for every gene in the default top 25/50/100, including unstable candidates; 8/17/41 survive every configuration. Whole-universe correlation and exact shared top-K counts are reported alongside the earlier conditional Spearman statistic on overlapping top-100 genes. The latter has mean 0.8548 and 16/24 above 0.85; it must not be described as full-ranking correlation. Each of six leave-one-layer-out analyses renormalizes the available remaining weights. Single-layer baselines use observed genes only, with their own denominators; no NULL-to-zero substitution is made. Default top 100 recovery is Usher 2/SYSCILIA 1, compared with animal-only 4/11. Literature-only recovers 2/9 Usher and 21/28 SYSCILIA controls in the top 100 (medians 99.13%/99.78%), whereas animal-only recovers 4/9 and 11/28. The integrated score has not been shown to outperform the strongest layer for either internal group.

S5. Frozen disease-comparator analysis and review-prompted diagnostics

The comparator protocol was committed at 0ae2c3d, and roster at 61abb61, before score-outcome inspection. Sources were the exact 2026-10-06 ClinGen validity export, exact HGNC live export and HPO v2026-09-01; SHA-256, download completion times and URLs are in the comparator source manifest in Additional file 11. The current revision distributes the two otherwise mutable live exports. Require Strong/Definitive validity, an Approved unambiguous HGNC Ensembl/Entrez mapping to the retained universe and HPO annotations. Exclude original controls and assertions/panels matching auditory, retinal or ciliary terms; exclude all descendants of the ten frozen HPO hearing, eye, motile-cilium, mucociliary and situs roots listed in the manifest. Reasons overlap and are not disjoint counts.

The eligibility pool contains 420 genes, of which 418 have non-NULL publication and GO counts. Matching log1p-transforms both raw counts and divides by their SD in the combined complete comparator/control population. Maximum-cardinality/minimum-squared-distance assignment supplies three unique matches per control, with no reuse and each absolute coordinate difference ≤0.75 SD (observed maximum 0.70945). All 37 controls matched: 27 Usher and 84 SYSCILIA comparators. Within-control averaging gives every control equal weight. Covariate SD differences after matching are publication/GO +0.0093/+0.0105 for Usher and −0.0412/+0.0385 for SYSCILIA. This balances two measured nuisance quantities rather than full research intensity or every layer.

Default median percentiles are Usher 97.77 versus matches 64.21, and SYSCILIA 91.93 versus matches 60.30. Default wins are 24/27 and 72/84; no pair is unranked. There are zero HIGH genes among the matches, but 110/111 already fail the raw threshold. The broader pool retains three HIGH genes, CRB2/ATP1A1/DYNC1H1. They were not removed after outcomes. A subsequent, non-systematic primary-source check found pre-existing sensory biology: CRB2 retinal degeneration

(https://pmc.ncbi.nlm.nih.gov/articles/PMC6747345/), ATP1A1 cochlear barrier (https://pmc.ncbi.nlm.nih.gov/articles/PMC3031570/) and DYNC1H1 photoreceptors (https://pmc.ncbi.nlm.nih.gov/articles/PMC7951903/). These explain residual overlap rather than validate novel Usher associations. Neither the frozen pool nor its remaining genes are verified biological negatives.

The additional post-outcome diagnostics follow the sensitivity protocol in Additional file 11. No production score, threshold, matching caliper or roster changed. Common-observed-layer scores compare each pair using only shared non-NULL layers, retaining original weights over those layers; they are pair-dependent means, not a new genome-wide ranking. All six single omissions and joint animal/literature omission use the original 111 pairs without rematching. A tie contributes 0.5 and an unranked pair stays missing. Every pair was ranked and no tie occurred. Pair means and equally weighted control means agree because every control has three matches.

Pair scheme	Usher wins / 27	SYSCILIA wins / 84
Default	24	72
Common observed layers	23	75
Omit constraint	25	76
Omit expression	27	78
Omit annotation	24	72
Omit localization	23	73
Omit animal	21	64
Omit literature	18	55
Omit animal and literature	15	48

Controls/comparators have different source coverage. Usher controls: seven five-layer/two six-layer, localization observed 2/9; their 27 matches: one four-layer/ten five-layer/16 six-layer, localization 17/27. SYSCILIA controls: two five-layer/26 six-layer, localization 28/28; their 84 matches: three four-layer/25 five-layer/56 six-layer, localization 61/84. Common-layer results limit one missingness explanation but do not remove ascertainment, shared curation or eligibility–feature overlap. Joint omission reduces concordance to 55.6%/57.1%. Full paired scores, denominators and source-observation balance are in Additional file 11.

S6. Retrospective temporal evaluation

S6.1 Preserved restricted pilot

The cutoff protocol (temporal_extension/protocol.md, commit 377eae3) preceded rank inspection. Three manually ascertained examples have later initial human associations located in the audit: CEP162 (2023 journal report; a subsequently located preprint was public on 23 November 2021), CFAP20 (initial candidate March 2022, subsequent independent November 2022 report) and LRRC45 (putative October 2021 report, later stronger evidence). Pre-cutoff ciliary biology was allowed; they are not a newly discovered Usher-positive cohort. Candidate and causal reports are distinguished. TMEM218 was excluded after primary full-text publication date 2020-11-21 contradicted apparent 2021 indexing. Other known pre-cutoff associations were excluded. Ascertainment was non-exhaustive.

The historical population uses Approved protein-coding HGNC entries in the 2020-10-01 archive with unique Ensembl mappings (19,167; two ambiguous rows excluded). Sources actually reconstructed were gnomAD 2.1.1 gene LOEUF, GO 2020-12-08 human GAF, HPA v20 normal-tissue ordinal protein and subcellular tables. Positive GO terms exclude NOT annotations and map through historical UniProt IDs, then unambiguous archived symbols. Annotation retains only its 0.5 GO component with missing UniProt/pathway subcomponents contributing zero, unlike wholly unavailable animal/literature layers omitted from the denominator. HPA ordinal data receive no quantitative Tau. Undated present-day compendia and modern identifiers/scores were not imported. Four layer weights remain 0.20/0.20/0.15/0.15, renormalized per observed

gene. All-gene components, six ranking schemes and case traces are preserved; no historical HIGH tiers were assigned.

Of 19,033 scored genes, default ranks for CEP162/CFAP20/LRRC45 were 8,289/11,004/253. Equal-weight ranks were 8,460/11,327/161; no default case entered top 100. LRRC45 localization was tied at the maximum with 77 genes; its favorable localization rank does not establish integrated success. All three lacked accepted retinal HPA observations. CEP162 protein categories were cerebellum/testis/fallopian 2/2/2; LRRC45 3/2/3; CFAP20 lacked all four panel observations. Ordinal categories are not quantitative fold changes.

Accepted HPA protein coverage in the historical population was retina 107 genes (all positive), cerebellum 10,894 (7,061 positive/3,833 observed zero), testis 10,917 (8,836/2,081), fallopian tube 10,898 (7,309/3,589). Top25/50/100 contained 10/11/11 one-layer genes. Historical top 100 comprised 11 one-layer, ten two-layer, 67 three-layer and 12 four-layer genes; 83 lacked localization, 22 expression, 12 annotation and three constraint. Missingness counts overlap.

This pilot is an archive/scoring diagnostic, not a complete historical six-family evaluation. Its restricted source contract and numerical outputs remain unchanged; the expanded study below uses separate sources and a separately fixed cohort. Animal and literature layers were not reconstructed in the pilot, and modern Census data cannot be assumed available at the cutoff. Manual case selection, partial layers, changed measurement coverage and sparse means prevent estimation of novel-gene sensitivity or independent HIGH specificity. Acquisition details remain in the preserved pilot audit.

S6.2 Expanded ascertainment and pre-outcome specification

The evidence cutoff was 31 December 2020, a common boundary for available dated archives, with clinical literature searched through 6 October 2026. The adopted RetNet catalog and ten PanelApp panels supplied 2,100 domain/source records and 1,081 gene/locus keys. Panels were 307, 126, 543, 724, 722, 178, 150, 725, 550 and 726, covering retinal, hearing and ciliary domains. Current and located pre-cutoff panel versions were retained with URLs/hashes. Reference metadata, gene/alias searches and primary reports were used to identify candidate later associations and check pre-cutoff reports. All 682 non-historical-green keys received complete broad searches across supported publication sources, including preprints: 57,153 records after full cursor pagination. Four ordinary-language aliases (ALL, LARGE, GAP and AT-1) were removed from the search aliases before outcome calculation; original attempts remain recorded. Titles/abstracts, references and full text were used for clinical review, but inaccessible reports and nomination filters can miss cases.

The target comprised inherited retinal/optic or hearing disease, or explicitly supported primary/motile ciliopathy, including ciliary renal, skeletal, hepatorenal, or laterality disorders. A broader sensory phenotype in a systemic disease required qualifying sensory genotype/phenotype evidence in at least two unrelated families. Isolated cataract, anterior-eye malformation or generic non-ciliary airway/renal disease did not automatically qualify. A first proposed target association had to occur after the cutoff; stronger evidence had to accumulate by the ascertainment date. The earliest located public human report included online publication, preprints and public theses rather than issue year, index date, catalog entry date or thesis award date. An earlier target candidate or modifier subsequently confirmed was maturation, not a new target association.

Principal clinical evidence required at least two unrelated target-phenotype families, consistent inheritance/segregation, and either functional support or independent target replication in a new family. Functional evidence required a variant effect, disease-related patient-cell/biochemical defect, relevant perturbation or rescue; healthy expression/localization or in silico predictions alone were insufficient. Source-specific family counts were distinguished from all-phenotype family totals. Material contrary human segregation or alternative diagnoses placed contested reports in the exploratory group. Existing Usher/SYSCILIA development controls and the three previously inspected pilot genes were excluded from principal recovery.

Clinical novelty was classified as A (no earlier causal human germline disease report located in the adopted audit), B (first target-domain report after other human disease evidence), phenotype expansion, maturation or unresolved evidence. A is an audit classification, not proof of universal literature absence. Prior somatic cancer/common-variant risk associations were documented separately. A/B were combined for principal first-target recovery and reported separately; B was not described as first-ever disease-gene discovery. Cases were provisionally selected through AI-assisted clinical/date evidence review without using newly calculated scores or historical source coverage to favor inclusion. Modern production outputs were accessible, so selection was not blinded. After the frozen analysis, human author review examined primary reports for all 45 cases, including reported families/pedigrees, phenotypes, inheritance, functional mechanisms and publication chronology. S6.6 records representative findings and unresolved family-evidence concerns. The original AI-assisted ascertainment and outcomes are retained; this later review is not blinded pre-outcome adjudication.

The final roster contains 1,092 unique keys, including 11 appended development controls absent from the catalogs: 45 principal strong cases, 18 candidates, three unresolved-novelty cases, 68 maturation records, six phenotype expansions, four scope comparisons, 50 development references and three pilot references. The remaining 358 historical curated records requiring target confirmation, 96 historical RetNet records requiring confirmation and 441 records without accepted later nomination retain clinical uncertainty. A closed selection decision is not completed clinical adjudication; these 895 entries are neither verified negatives nor an exhaustive clinically resolved background. Out-of-catalog associations were not added ad hoc. Recovery therefore applies to this ascertained positive cohort.

First-publicity intervals were retained for CREB3 (2024-05-01 to 2025-07-17), MRPL49 (2022-01-01 to 2024-10-11) and TMEM72 (2021-01-01 to 2022-04-30), where identified thesis/conference evidence lacked an exact verified public day. These intervals lie wholly after the cutoff. The subsequent publication-version audit also retained earliest-day uncertainty for ATP11A (S6.6). DAP3, LETM1 and TBX2 remained unresolved-novelty cases outside the principal cohort. The clinical ledger records dates, citations, target families, inheritance and functional/replication support.

The protocol, source inventory, implementation and roster were committed at bd6bb2f before any expanded composite/rank/HIGH/recall calculation. This was an internal pre-outcome specification, not prospective registration. A guarded initial reconstruction stopped at a GTEx duplicate-ID check: version removal had conflated 44 canonical and _PAR_Y pairs. The correction removed version digits while preserving the Y suffix as a distinct key, and was committed with its regression test and amended dependency manifest at 53da295 before the successful ranking run. No cohort, source, weights or gate changed. The original freeze and failed-attempt record remain preserved. Protocol/evidence design and production calibration occurred after 2020, so even a pre-outcome expanded specification cannot make this a prospectively developed predictor.

S6.3 Historical sources and changes from production

The archived HGNC 1 October 2020 Approved protein-coding set with unique nonempty Ensembl mappings contains 19,167 genes; two ambiguous identifier rows were excluded. Identifiers were resolved through frozen HGNC/Entrez/Ensembl keys and unique archived aliases. The production score table was not imported into historical evidence construction. Source dates differ within the cutoff; the reconstruction does not represent complete evidence accumulated through 31 December 2020.

Family	Adopted historical source	Difference or limitation
Constraint	gnomAD v2.1.1 gene LOEUF	Inverse min-max transform over the historical universe; older release than production
Expression	HPA v20 ordinal protein, GTEx v8 and original Menon 2019 retinal data [45]	GTEx has no retina or cochlear hair cells; retinal matrices replace modern Census

Family	Adopted historical source	Difference or limitation
Annotation	GO human GAF 8 December 2020	Available GO term component only; UniProt annotation-score/pathway channels not recovered, maximum family score 0.5
Localization	HPA v20 subcellular observations	Undated embedded compendia excluded; absent observations NULL and recorded non-ciliary location zero
Animal	MGI human/mouse links/phenotypes 2 February 2020, MP vocabulary 29 January 2019, IMPC release 11 and ZFIN 30 December 2020	Native curated ortholog links with unit weight; historical HCOP confidence votes not recovered
Literature	June 2020 gene2pubmed plus six date-bounded context PMID sets	Modern title/abstract text; residual later text-revision uncertainty

GTEx v8 was publicly released 26 August 2019. Its GCT contains 56,200 rows across 54 tissues, including 44 preserved _PAR_Y keys; only canonical historical identifiers join the corresponding canonical rows. Quantitative restricted-panel Tau uses available GTEx cerebellum, testis and fallopian-tube TPM columns. Downloaded HPA v20 RNA contained neither retina nor cerebellum, so its background-only TPM was excluded rather than mislabeled as nTPM or used as a target contrast. Ordinal HPA protein categories use the existing within-tissue percentile fallback, not raw quantitative ratios.

GSE137537 (10X; public 21 September 2019) and GSE137846 (Seq-Well; public 23 September 2019) supply original raw-count matrices, gene features and author cell labels. All originally labeled Rods/Cones were selected without new clustering: 9,356 and 847 cells. The Seq-Well header omitted the barcode column, which was restored explicitly after checking row widths, barcodes and matrix dimensions. Features map by exact archived symbol followed by unique historical previous/alias symbol; duplicate features assigned to one gene are summed. Per-gene raw UMI means include zero counts. Pooling sums counts and cell denominators only across platforms where the gene is an observed feature; absence from both feature lists is NULL. Counts are independently percentile-normalized within the expression transform and are not claimed to be TPM or a historical Census snapshot. No direct hair-cell component was introduced.

GO positive unique terms exclude NOT annotations; archived HGNC UniProt accessions and unambiguous symbol fallback provide mappings. The production annotation formula retains its 0.5 GO component, with unavailable UniProt/pathway subcomponents contributing zero and wholly absent GO yielding family-level NULL. A named GO-rescale sensitivity divides available annotation by 0.5; the primary family was not rescaled after outcomes.

Mouse phenotype terms are unioned across native mapped orthologs and MGI/IMPC, deduplicated by relevant term name. Historical IMPC names supplement the older MP vocabulary. Recognized positives with unknown terms yield a documented lower bound; unresolved-label-only channels yield NULL. ZFIN abnormal records require an unambiguous native gene ID, with compound/multigene assignments excluded; native gene/structure/name tuples are deduplicated across publication/figure rows before human-ortholog union. The 30 December archive avoids the 31 December file's 1 January 2021 modification boundary. No new p-value filter was imposed. A mapped, interpretable channel without relevant terms supplies observed zero, not a biological negative. The primary animal score is (0.4*mouse_positive + 0.3*zfish_positive)*log2(combined_relevant_count+1)/log2(global_maximum+1) with native-link unit weights; no channel mapping yields NULL.

Literature uses complete human Entrez/PMID links from the June 2020 archive. Europe PMC [44] supplied clinical searches and historical context queries. All six contexts require MEDLINE source (SRC:MED), FIRST_PDATE (earliest electronic/print publication) and FIRST_IDATE (first Europe PMC search indexing) <=2020-12-31, using title/abstract text rather than modern MeSH. Clinical searches can include preprints separately; these context sets cannot. Unique context-set sizes were cilia 32,887, sensory 230,790, cytoskeleton

171,720, polarity 17,767, high-throughput 180,886 and direct-experimental 11,935; each acquisition completed pagination with IDs matching returned hits. Counts, same-PMID direct evidence, tier precedence and publication normalization follow S3. A mapped Entrez gene without links is observed zero; absent identifiers would remain NULL. Date filters do not recreate historical text, so the five-layer no-literature and six-layer augmented versions are co-primary. Held-out association PMIDs were checked against corresponding archived gene links; none leaked.

S6.4 Comparisons, endpoints and HIGH convention

The two co-primary schemes use production weights 0.20/0.20/0.15/0.15/0.15/0.15, with literature weight zero for five layers and available-family renormalization. Observed zero contributes to the denominator; NULL does not. The 21 additional schemes comprise equal weights, six individual families, six single weight omissions, joint animal/literature omission, uniform animal multipliers 0.7/0.4, unique-native-link animal evidence, no-photoreceptor and each-platform expression, and available-GO rescaling. No parameters were fitted using principal case recovery.

Every scheme reports top 25/50/100/500/1000 and top 1/5/10% (ceiling of fraction times its scored universe). Deterministic descending-score positions break ties by ascending Ensembl ID. Statistical minimum/maximum tie ranks and ascending average score percentiles 100*(rank-1)/(n-1) are separately saved. Tie-inclusive recovery accepts a tie starting within the cutoff; the conservative whole-tie rule requires its maximum rank within the cutoff. A NULL or absent-universe case fails all recovery rules.

Three denominators are reported: eligible historical-universe cases including unscored failures, all clinically eligible cases including absent-universe failures, and scored-only cases for descriptive comparison. The principal denominators are 41 and 45: all 41 historical cases are scored, while RNU4-2/RNU6-2/RNU6-8/RNU6-9 are noncoding and outside the fixed protein-coding universe. Novelty, target-domain and mechanism strata retain original labels; target domains overlap and cannot be summed. Candidate, unresolved-novelty, expansion/maturation and development/pilot results are separate. Wilson 95% bounds are descriptive within this nonrandom cohort, not population validation intervals. No clinical negative cohort, specificity, precision or false-positive rate was estimated.

Historical HIGH requires score>=0.7, at least three observed families, including zero values, with weight > 0, and either one of the five direct HPA compartment flags or animal score at least the historical positive-score Q75. Q75 uses Polars nearest interpolation, matching production; it was 0.13475306857428798 in the primary historical animal vector. Animal-source variants recompute it over the full historical universe without case labels. Uniform scaling preserves the Q75 gate but can alter composite scores. Count>=4 and the original fixed production animal threshold 0.12765713253595054 are named sensitivities with cross-version scale limitations.

Leave-one-family analyses are weight omissions: evidence tables and both gate routes remain available, while count uses positively weighted observed families. They are not complete removal of that source from HIGH. Animal-source variants change gate inputs; localization-weight omission retains direct HPA flags. A single-family baseline has only one active family and cannot pass the count criterion. The gate architecture remains development-control-calibrated and is not revalidated as disease probability by later-case recovery.

S6.5 Outcomes and validation

The five-/six-layer versions score 19,157/19,167 historical genes. Principal top 100/HIGH recovery is 0/41 (0/45 end-to-end), top 1,000 is 2/41 versus 3/41, and top 10% is 5/41 versus 6/41. Median ascending percentiles are 46.23/44.95. All 41 have at least four observed weighted families but none score>=0.7. Thus the absence of HIGH recovery is not explained solely by the gate. Constraint, expression and GO are positive for all 41; localization is positive/zero/NULL for 2/29/10, animal positive/zero for 13/28, and literature

positive/zero for 25/16. Across the whole genome HIGH counts are zero/five, illustrating source-scale differences as well as limited later-case recovery.

No principal case reaches HIGH in any of 23 schemes. Single localization recovers SAXO6 at deterministic top 100 (position 19, tied 1-77); localization-weight omission recovers PLCG1 at 90 without a boundary tie. Both satisfy the whole-tie rule, but neither replaces the co-primary results. Five-versus-six ranking correlation is 0.870708 across 19,157 shared scored genes; full top 100 sets overlap by 57. These findings do not establish superiority, broad novel-gene recovery or a formal below-chance result.

All unchanged historical constraint/GO/localization components reproduce the preserved pilot. All nine output TSVs and the manifest were byte-identical in a separate clean reconstruction. At that stage, the Python 3.13.1/macOS-arm64 environment passed 354 tests, with 17 existing warnings. A separate internal OpenAI gpt-6-astra/high AI-agent audit (not human clinical adjudication or external validation) rebuilt the raw sources and all case/summary arithmetic; agreement supports implementation consistency, not external clinical validity. Original production and baseline hashes remained unchanged. Additional file 12 supplies the final protocol, clinical reference ledger, source inventory, environment lock, all outputs and reconstruction/evaluation scripts. Large raw archives and clinical search/full-text caches remain outside the distributed bundle; hashes and original acquisition records identify them.

S6.6 Subsequent human author review

A human author examined primary reports for all 45 cases after the frozen temporal analysis, including reported family/pedigree information, phenotype, inheritance, functional mechanisms and publication dates. The review distinguished the number of individuals from unrelated families and separated the target phenotype from earlier disorders associated with the same gene. It was a source and interpretation check, not a new blinded prospective benchmark.

Three reports prompted particular family-evidence concerns. The initial *PDIA6* paper described one affected individual from one family [57]. The earlier *VWA8* developmental-disorder paper described nine individuals from one family [58], whereas the temporal target was a later retinal association. For *PLCG1*, a single-family immune-disorder report [59] was distinguished from a later multisystem report with auditory/ocular involvement and four families [60]. The frozen criterion requires at least two unrelated target-phenotype families cumulatively across eligible reports, not necessarily within each paper or displayed pedigree. The frozen screening ledger also cites later PDIA6 reports (PMIDs 35856135, 40974269), recording three families overall and a conservative lower bound of two target families; the VWA8 retinal reports (PMIDs 37012052, 40638000) give a target-family lower bound of two, and PLCG1 has a conservative target-family lower bound of three. These are the retained screening classifications, not new human inclusion decisions. Eligibility remains under review; the original cohort and recovery estimates are retained.

The review also distinguished *CRLS1* mitochondrial cardiolipin deficiency with sensory manifestations [61], *CTNND1* retinal vascular disease [62], *DHRSX* dolichol-biosynthesis/glycosylation disease [63], and spliceosomal RNA-associated retinal disease involving *RNU4-2* and RNU6 genes [64]. Sensory manifestations alone do not establish a ciliary mechanism. The temporal scope intentionally includes inherited retinal/optic disease, hearing impairment and explicitly supported ciliopathies, rather than only mechanistically established ciliopathies.

Publication chronology was checked against the supplied reports and bibliographic records, distinguishing first public target evidence, preprints/early-online versions, acceptance and issue dates. Eligibility uses a supported first-publicity interval wholly after 31 December 2020; an exact day is unnecessary when the interval does not cross the cutoff. Most discrepancies concerned publication stages. For example, TMEM72 conference/abstract evidence in 2021–2022 predates the full paper’s 2025 online/2026 issue dates. Exact earliest-day uncertainty remains documented for ATP11A, CREB3, MRPL49 and TMEM72. No newly confirmed pre-cutoff target association was found in this bounded audit. The original dated roster and numerical results remain preserved.

Additional file 12 includes this review summary alongside the frozen analysis records; the complete working author-review form is not an additional submission attachment.

S7. Numerical replay, provenance and submission-file checksums

The revision freezes the original 20,081-label score table plus tissue-expression, annotation and literature tables as canonical sorted TSVs with schemas, explicit NULL tokens and SHA-256 hashes. A rebuilt task-owned DuckDB must reproduce all four canonical table hashes before analyses proceed. This derived bundle includes restored donor-derived annotation/animal fields and enables numerical replay, not reconstruction of every original raw source. Original unrecorded retrieval dates remain unknown. Submission manuscript text is preserved separately for numerical baseline checks, so editing the revision does not invalidate replay.

revision/major_revision_20261006/reproducibility/README.md provides exact commands, the tested dependency lock, source inventory and limits. Control/robustness diagnostics, frozen matched-comparator outcomes, the restricted historical reconstruction and post-outcome sensitivities were replayed in a clean Python 3.13.1 macOS arm64 environment with pinned versions. Dependency compatibility was checked; the current suite, including the GO extension, passed 362 tests with 17 existing warnings. Numerical TSV outputs are compared by bytes rather than merely matching rounded summaries. Figure sampling uses 500 of 20,016 non-control scored labels without replacement, NumPy default_rng(42), ordered by retained Ensembl ID; exported sampled IDs permit checking the display population.

The submission checksum issue is real: the original Additional file 10 entry identifies repository Markdown expression_shortlist_report.md, not the submitted TXT bytes. The Markdown SHA-256 is e046dbcd6b9eebd8190ad75a9c1cc4d0f0396cd887e48b6558f7d25633a1967f; submitted Additional file 6 TXT is 13fc6ddd5f3ef1f2b27d4812f70944a61824a394f3ec4e6f54619bc883ce9461. The revision records source and converted artifact separately. The revised upload set was assembled under submission/bmc_bioinformatics_major_revision_20261020_v2/. Additional file 10 hashes the actual delivered additional files 1-9 and 11-13, including the converted Supplementary Methods and expression-shortlist TXT files, without self-hashing. The separate package manifest records manuscript/response DOCX and PDF hashes plus their source hashes and numerical checkpoints. Rendered page checks and page-based response locations apply to this generated version. No DOI-bearing archive has been minted; exact Git analysis checkpoints and artifact hashes are provided instead. Source-file hashes do not substitute for converted-file hashes.

S8. Contemporary GO classification of the temporal cases

This post-hoc extension describes the same 45 temporal cases using contemporary human GO annotations, without changing frozen case labels, scores or outcomes. Protein and RNA GAF files generated on 11 October 2026 were mapped through stable HGNC keys, explicit UniProt/RNAcentral accessions and unambiguous symbols. GO release 27 September 2026 was pinned. Version-matched QuickGO is_a/part_of ancestors were restricted within the original BP/MF/CC aspect; the separately pinned official GO editor snapshot supplied generic-slim membership and alternate/obsolete-ID resolution. NOT, ND/root-only and unresolved obsolete annotations were excluded. Unique gene–term counts retained source evidence/reference records. Non-IEA support includes curated inference and is not synonymous with experimental support. Shared RNAcentral mappings were disclosed. Categories overlap, and absent annotation is missing evidence rather than absent function.

All 45 cases had usable GO annotations, with BP/MF/CC coverage of 41/42/44. The four noncoding RNU genes remained in the descriptive 45-case overview, with IEA-only annotations, but outside the original coding-score/enrichment universe. Current GO categories were not treated as information available in 2020. The primary enrichment query comprised the unchanged 41 coding cases and reference comprised the 19,167 historical coding genes, conditioned on usable annotation within each aspect. Annotated query/reference denominators were BP 37/16,831, MF 38/17,541 and CC 40/18,083.

One-sided hypergeometric over-representation tests included every propagated non-root term with at least five reference genes inside and outside the term, without selecting by case overlap. Zero-overlap terms remained in the family. BH FDR pooled BP/MF/CC terms within each analysis. Two secondary analyses applied the same rules after excluding IEA in query/reference, or restricting reference to the 1,032 catalog-sourced genes in the historical coding universe. Catalog reference genes are not clinical negatives. Each analysis had its own multiple-testing family.

There were 11,388 primary, 10,459 non-IEA and 2,750 catalog-background tests. None passed FDR < 0.05; minimum adjusted values were 1.0, 0.3531 and 1.0, respectively. Overlapping descriptive categories included cilium and mitochondrion (eight cases each), cilium organization (four), and mRNA metabolic process (five, including four RNU genes supported only by IEA). This composition is consistent with a broad sensory-disease target and cannot establish ciliopathy mechanisms or explain ranking failure causally. The 140 generic-slim categories yielded 85 nonempty categories and 341 memberships.

Additional file 13 contains the readable seven-sheet workbook, complete source annotations and enrichment tables, protocol/freeze/amendments, source manifests and numerical verification. Figure S1 shows the 12 most frequent nonempty generic-slim categories per aspect, with GO-ID tie-breaking and all 45 cases as denominator; bars distinguish non-IEA support from category-level IEA-only support. The full gene/category tables retain categories outside the figure. Independent Fisher-exact/SciPy-BH recalculation agrees for all 24,597 tests, all frozen case-percentile joins are exact, and protected baseline hashes are unchanged. This post-hoc functional description is not an additional temporally held-out predictive endpoint.

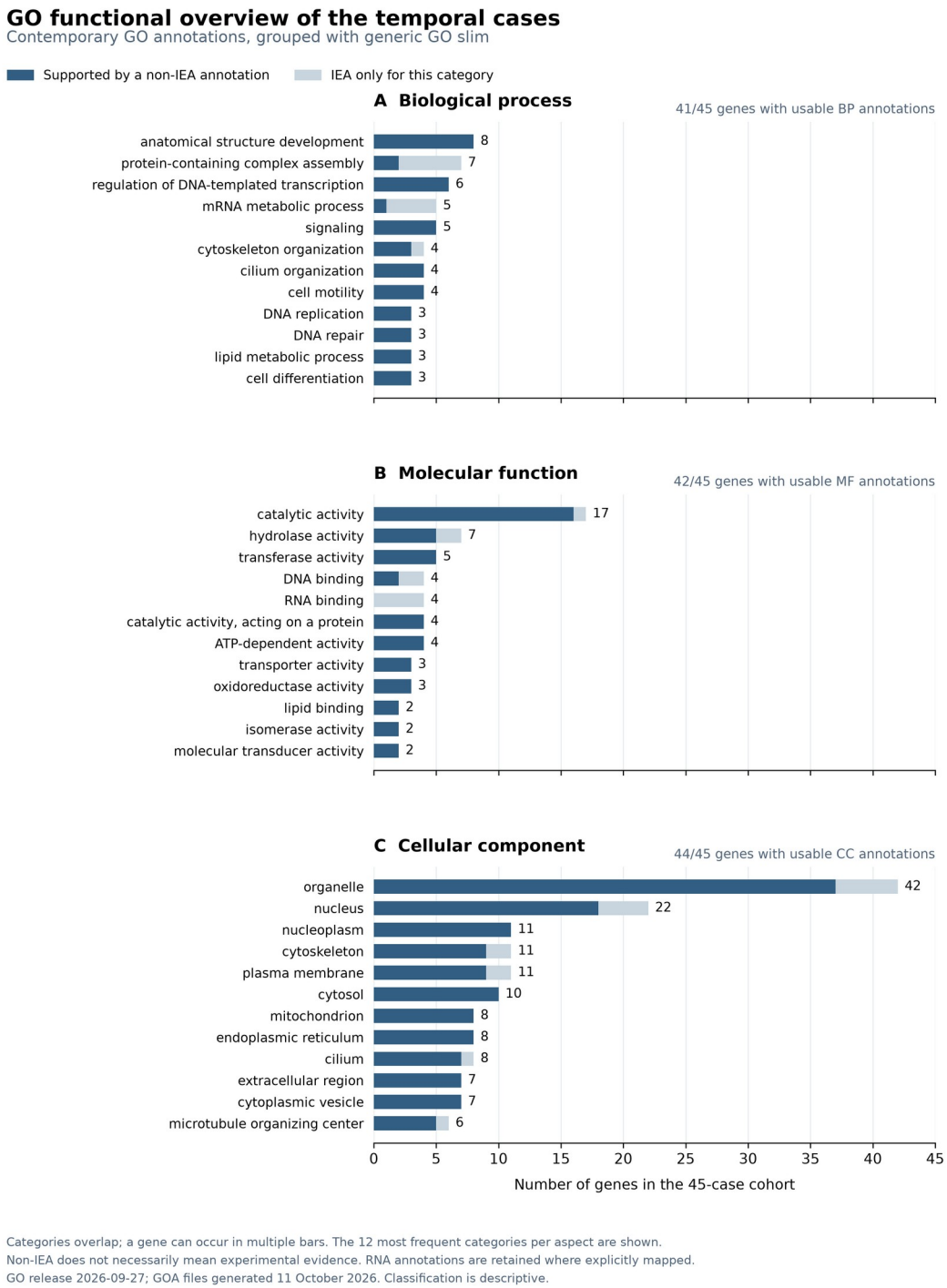


Figure S1. Contemporary GO functional overview of the 45 temporal cases. Panels show biological process, molecular function and cellular component categories. Categories overlap; the plotted counts are not a partition of the cohort. Dark bars indicate support from at least one non-IEA annotation; light portions indicate IEA-only support for that category. Non-IEA does not necessarily indicate experimental evidence. Four explicitly mapped noncoding RNAs are retained in the overview, with shared sequence-mapping limits disclosed in the workbook. GO release 27 September 2026 and GOA files generated 11 October 2026 describe contemporary function and do not enter historical scoring. No enrichment analysis passed the specified FDR threshold.